Leyla Yanik

Doctoral trainee · EPSRC RAINZ CDT

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PROFILE

Psychologist turned roboticist. Background in experimental psychology (EEG, eye-tracking, developmental and memory research) and an MSc in applied AI, where I built and evaluated a YOLO11 toy detector for a child-language robot. Now in first-year robotics training with the RAINZ CDT, before a PhD at Glasgow on privacy-preserving robotic inspection of net-zero infrastructure.

EDUCATION & TRAINING

Sep 2026–present
Doctoral training, EPSRC RAINZ CDT
Year 1 robotics training, University of Manchester: Cognitive Robotics and Computer Vision; Robotic Systems; Software for Robotics; Robotic Systems Design Project.
PhD in Autonomous Robotics, University of Glasgow, from September 2027. Project: *PRINZ: Privacy-Preserving Robotic Inspection of Net-Zero Infrastructure*. Supervisors: Prof. Mohamed Khamis, Dr Euan McGookin.

2025–2026
MSc Applied Artificial Intelligence and User Experience, Abertay University · **Distinction** (GPA 4.5/4.5)
Dissertation (98%): *Domain-Specific Toy Object Detection for Child–Robot Interaction: Developing and Evaluating a YOLO11-Based Perception System for WordMate*. Supervisors: Prof. Vera Kempe, Dr Marwan Mohammed.

2021–2025
BSc Psychology, University of Edinburgh · **First-Class Honours**
Dissertation (85%): *Forgetting of relational and conjunctive bindings*.

SELECTED PROJECTS

2026–27
Autonomous object retrieval robot (MSc Robotics team design project)
Five-person team building a Leo Rover with a robotic arm that searches for coloured objects, retrieves them and sorts them into matching bins, using ROS 2 and an Intel RealSense RGB-D camera. One of the team's coordinators; set up and lead the team's GitHub repository and documentation.

2026
WordMate: toy recognition for child–robot interaction (MSc dissertation)
Collected and hand-annotated 1,176 RGB images (GoPro, Intel RealSense D435i), audited labels and consolidated a 43-class taxonomy; fine-tuned YOLO11s with five-fold cross-validation. Mean mAP@50 0.901 (SD 0.011); correcting labels raised recall from 0.780 to 0.846. Tested live within the WordMate system.

Jan–May 2026
Curriculum learning via multi-armed bandits (CMP511 AI & Machine Learning)
Teacher–student framework in which UCB, discounted UCB and Thompson-sampling teachers choose arithmetic tasks for GPT-2/DistilGPT-2; compared with uniform and easy-to-hard baselines across seeds, with reward and capacity ablations. Code: github.com/lyanik/curriculum-llm

2024–2025
Forgetting of visual feature bindings (BSc dissertation)
Two-session memory experiment (N = 82) comparing conjunctive and relational bindings over one and seven days; analysis in R.

RESEARCH EXPERIENCE

2026–present
WordMate research volunteering: data collection, annotation and interaction activities for AI-supported child language learning.

2024–2025
Psycholinguistics: eye-tracking data collection and archiving, literature reviews, stimuli and experiment design for sentence prediction and comprehension studies.

2024–2025
Developmental psychology: behavioural, EEG and eye-tracking data collection.

2022–2023 **Polysemy research:** coding naturalistic caregiver–child conversations.

OTHER EXPERIENCE

2022–2025 **Information Services Helpdesk Assistant**, University of Edinburgh. Supported access to information services and contributed to UX work, including student interviews.

SKILLS

Robotics & systems	ROS 2, Linux (Ubuntu), Intel RealSense depth camera, GoPro data capture
Code & tools	Python, PyTorch, R, SQL, SPSS, Jupyter, Google Colab, Git/GitHub, Markdown
AI & perception	Computer vision, YOLO (Ultralytics), Hugging Face Transformers, multi-armed bandits, cross-validation, dataset annotation (Roboflow)
Research methods	Experimental design, statistical analysis, EyeLink eye-tracking, EEG, UX research, Gorilla, Praat
Languages	English (C2), German (C2), Turkish (B1)

AWARDS, ACTIVITY & DEVELOPMENT

- EPSRC RAINZ CDT Doctoral Studentship (TechFirst Doctoral Support), 2026
- Data Lab Academy Scholarship, 2025–2026 · The Data Lab Masters Programme, 2026 · DataFest 2026 (8 CPD hours)
- Presenter, BPS Undergraduate Conference, 2025
- Organising committee, Architectures and Mechanisms for Language Processing (AMLaP), 2024
- Research output: Yanik, L. (2025). *Eye-tracking setup line drawing* [Figure]. Figshare, CC BY 4.0. doi:10.6084/m9.figshare.28761359